

TEJAS PHUTANE

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SUMMARY

Robotics and Computer Vision engineer with 3+ years of hands-on experience in designing and deploying vision systems, autonomous platforms, and AI solutions for industrial automation. Proven expertise in end-to-end system architecture, real-time perception pipelines, and production-grade implementation. Strong background in scaling complex projects involving hardware integration, software optimization, and technical team leadership. Published researcher and thought leader in robotics innovation with track record of delivering measurable business impact.

SKILLS

Expertise: Motion Planning, Trajectory Control, Gazebo, Isaac Sim, MoveIt, Sensor Fusion, Localization, Path Planning, Kinematic Analysis
Programming: C++ (Proficient), Python (Expert), C, MATLAB
Technical Tools: ROS2, Isaac Sim, Gazebo, RViz, Git, Docker, YOLOv11, Linux/UNIX, Deepstream, AWS
Hardware: Nvidia Jetson Xavier, Multi axis Motion Controller, Industrial Cameras

WORK EXPERIENCE

Wastefull Insights

Vadodara, India

Senior Robotics Engineer

Aug 2023 - Nov 2025

- Led **software architecture, perception pipeline** development, and technical team coordination for autonomous robotic systems. Managed end-to-end project delivery from design through production deployment.
- Architected **motion planning** and **control algorithms** for **3-axis gantry systems**, reducing inertial motion losses by 15% and improving system efficiency by 80%. Increased overall system speed and accuracy by 40% through **iterative algorithm refinement** and **trajectory optimization**.
- Coordinated with hardware vendors for camera, lighting, and motor integration; ensured seamless hardware-software synchronization.
- Built **robot-vision analysis framework** with database integration for continuous performance monitoring and optimization. Developed regression models for throughput prediction with 91% accuracy.
- Engineered custom multi-robot coordination algorithms for waste sorting facility processing 500+ objects/hour; optimized task allocation reducing idle time by 30% and increasing overall throughput by 25%

Robotics Engineer

Aug 2022 – Sept 2023

- Refactored codebase from Python to C++ for performance-critical modules, reducing computational overhead by 80% and enabling real-time processing.
- Established 2x faster and memory optimized multimodel YOLO pipeline for detection and accurate tracking of waste objects using Nvidia Deepstream framework

Engineering Services International (Science City, Robotics gallery)

Ahmedabad, India

Junior Robotics Engineer

Oct 2021 - Aug 2022

- Architected perception and navigation stacks for autonomous mobile robot platforms using ROS ecosystem. Designed and validated motion planning algorithms in Gazebo simulation environment, reducing development cycle by 60%.
- Implemented multi-modal sensor fusion combining camera and Lidar data with Kalman filtering for robust localization.
- Developed object detection pipeline using OpenCV and YOLOv3 for real-world robot vision applications.
- Developed and implemented fast and robust algorithms for Disc detection and trajectory estimation using opencv and ROS. Worked on Epson Scara robot and implemented defense and attack strategies based on trajectory and position of disc.

IN2PETA Services

Hyderabad, India

Computer Vision and ROS Intern

June 2021 - Oct 2021

- Developed an autonomous pick and place mobile robot for toys clutter environment pick and place
- Developed and tested object detection, high level planner, path planning, sensor fusion using ROS on Nvidia Jetson
- Developed computer vision solutions using OpenCV

EDUCATION

University of Mumbai, MCT's Rajiv Gandhi Institute of Technology
Bachelor of Engineering, Electronics & Telecommunication Engineering (CGPA: 7.04)

Mumbai, India
March 2021

TECHNICAL EXPERIENCE

Precision Manufacturing Vision QC System

October 2025

- Conducted reverse engineering of legacy control systems and PLC programming Developed high-performance defect detection model trained on 5000+ annotated samples
- Integrated hardware stack including industrial cameras, line-scan optics, and precision lighting Coordinated hardware procurement and mechanical system calibration with vendors
- Reduced manual inspection labor by 60% and improved defect detection consistency and established continuous deployment pipeline for model updates and performance validation

Multi-Stream Vision Analytics Platform

October 2025

- Technologies: Python, C++, YOLOv5, NVIDIA DeepStream, TensorRT, Kafka, TimescaleDB, PostgreSQL, Docker, Streamlit, REST APIs
- Real-time video processing system analyzing multiple RTSP feeds simultaneously
- YOLOv5-based object detection with secondary classification for SKU identification and counting
- End-to-end pipeline: NVIDIA DeepStream ingestion, TensorRT inference optimization, Kafka message streaming, TimescaleDB time-series storage
- Achieved 99%+ detection accuracy with sub-50ms latency per frame
- Integrated REST APIs for real-time analytics queries and dashboard visualization using Streamlit
- Deployed to production environment with continuous monitoring and model retraining pipeline

Waste Sorting Robot-Vision Analysis Framework

August 2025

- Built database-backed system logging all robot operations and vision detections
- Implemented analysis tools for performance metrics extraction and trend analysis
- Developed regression models predicting system throughput based on operational parameters
- Created visualization dashboards for real-time monitoring and historical performance review
- Enabled data-driven optimization decisions reducing cycle time by 15%

Autonomous Warehouse Management System (Eyantra Robotics Competition) | [LINK](#)

September 2020

- Developed autonomous warehouse management system using two UR5 robots in ROS Gazebo environment ,used Python nodes for controlling robot arms using MoveIt motion planning framework.
- Implemented IOT using MQTT protocol for communication between robots and visualized package delivery progress on google spreadsheets using Google App Scripting

CERTIFICATIONS

- Introduction to Robotics (NPTEL)
- Robotics Specialization (Coursera)
- Control of Mobile Robots (Coursera)

